

08 The Application Layer

08.02 DNS

In the Beginning

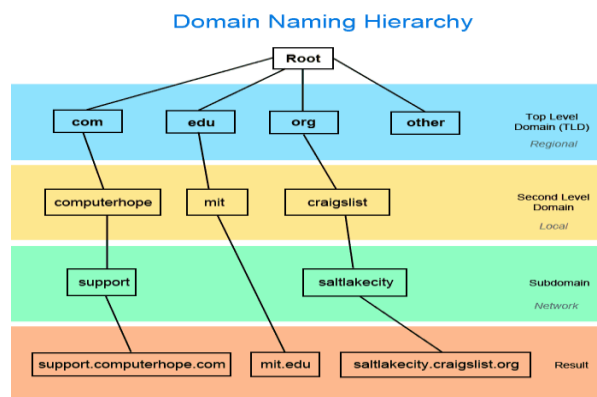
- The Internet was small
- Fewer than 100 hosts
- Everybody knew everybody else
- Centralized: host file distributed to everyone
- But it didn't scale

DNS – The Domain Name System

- We need to introduce ASCII names to identify hosts (separate names from IP addresses)
- DNS is a hierarchical denomination scheme
- RFC1035
- It is based on a distributed database system for its implementation
- To associate IP address to a name the application:
 - Calls a library procedure (resolver), e.g. `gethostbyname()`
- With IP address application can establish a TCP connection or send UDP packets

The DNS Name Space

- Hierarchical addressing (like postal address)
- It consists of the root directory, represented as a dot (.) and two top level domain hierarchies
 - organizational
 - geographical



Hierarchy of DNS Servers

Root DNS server

- Top level domain server
- Authoritative server

Each domain ISP/University/Company has a local DNS server

- When host makes a client DNS request, query is sent to local DNS which acts as proxy server

Each domain controls the allocation of the underlying domains

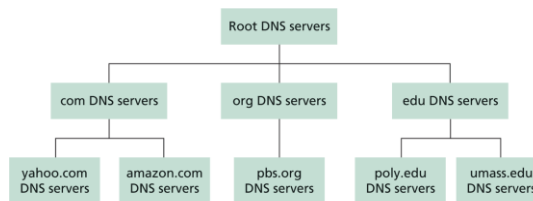


Figure 2.18 ♦ Portion of the hierarchy of DNS servers

DNS

- Internet root servers in 2004



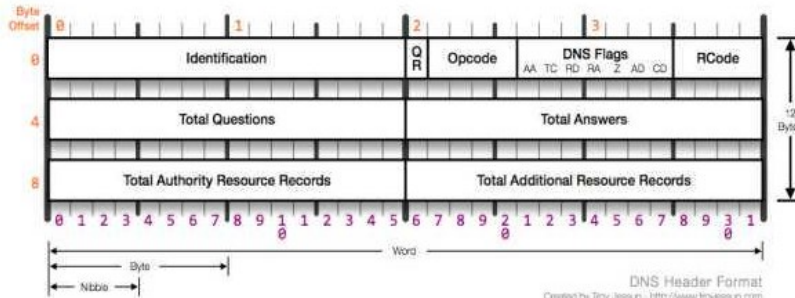
Copyright © 2005 Pearson Addison Wesley All rights reserved ♦ DNS root servers in 2004 (name, organization, location)

Name resolution steps

1. A DNS server is configured with an initial cache (so called hints) of the known addresses of the root name servers. The hint file is updated periodically in a trusted, authoritative way.
2. When a client makes a (recursive) request of that server, it services that request either through its cache (if it already had the answer from a previous lookup) or by performing the following steps on the client's behalf.
3. A query is made to one of the root servers to find the server authoritative for the top-level domain being requested.
4. An answer is received that points to the nameserver for that resource.
5. The server "walks the tree" from right to left, going from nameserver to nameserver, until the final step which returns the IP address of the host in question.
6. The IP address of the resource is then given to the client.

DNS Protocol

- The DNS protocol is relatively light (see the image above) and thus sits on top of UDP so that queries can happen quickly and without much overhead. Queries over 512 bytes, and certain heavier operations such as Zone Transfers, however, switch to using TCP so that delivery will not become an issue.



Protocol's fields

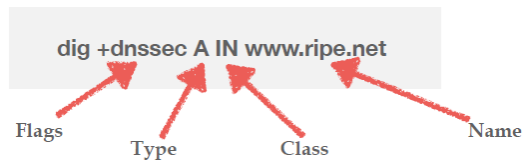
- **The Identifier:** a 16-bit ID field that matches requests and responses.
- **The Query/Response Flag:** a 4-bit field that designates whether the packet is a request or a response.
- **Opcode:** Specifies the type of message being carried. Options include: 0 for standard query, 1 for an inverse query (obsolete), 2 for server status, 3 is reserved and unused, 4 is notify message, and 5 is an update (used for Dynamic DNS).
- **AA:** This is a 1-bit field indicating an authoritative answer. The bit is set to 1 if it's authoritative, meaning that the server who gave the answer is authoritative for the domain in question. If it's set to 0 it's a non-authoritative answer.
- **TC:** A 1-bit field for truncation, yes or no. Usually indicates it was sent via UDP but was longer than 512 bytes.
- **RD:** A 1-bit field called "Recursion Desired", meaning that the client is asking the server to walk the tree on the client's behalf and just return the answer as opposed to telling it where to look next.
- **RA:** A 1-bit field called "Recursion Available", in which a DNS server tells a client that it support recursion or not.

Protocol's fields

- **Z:** Three reserved bits that are always set to zeroes.
- **RCode:** A 4-bit field that's set to zero in queries (because they're not responses) with the following options: 0 is no error, 1 is format error, 2 is server failure, 3 is name error, 4 is not implemented, 5 is refused, 6 the name exists but it shouldn't, 7 a resource record exists that shouldn't, a resource record that should exist doesn't, 9 the response is not authoritative, 10 the name in the response is not within the zone specified.
- **QDCount:** How many questions in the question section.
- **ANCount:** How many answers in the answer section.
- **NSCount:** How many resource records in the authority section.
- **ARCount:** How many resource records in the additional section.

DNS Query

- Every DNS Query consists of:
 - qname: a domain name (i.e. www.ripe.net)
 - qtype: A, AAAA, MX, CNAME, PTR, SRV, TXT, NS
 - qclass: IN, CH, HS (only IN used today)
 - Flags: QR, RD, EDNS Opt, DO, AD, etc.



Resource Records

- A set of *resource records* can be associated to each domain (a single host or top level domain)
- DNS associates domain names to resource records
- Resource records are obtained by resolver procedures
- Resource record:
 - Name TTL Class Type Value
- Name: the domain to which the record belongs to
- TTL: used to discriminate volatile and non-volatile information
- Class: IN for Internet information
- Type: specifies the record type (see next table)
- Value: a number, a name, an ascii string (semantic dependent on type of record)

Resource Records

- SOA: used to provide the primary information source on name server zone
- A: contains the IP address of some host
 - If a host has 2 interfaces→2 IP addresses → a resource record for each interface
- MX: specifies the name of host configured to accept mail for that specific domain
- NS: specifies the name server
- CNAME: allows to create alias
- PTR: used for reverse lookups
- HINFO: used to know type of machine and OS
- TXT: optionally used to allow arbitrary domain self-identification

Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone
A	IP address of a host	32-Bit integer
MX	Mail exchange	Priority, domain willing to accept e-mail
NS	Name Server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
HINFO	Host description	CPU and OS in ASCII
TXT	Text	Uninterpreted ASCII text

The principal DNS resource records types.

Resource Records

```
; Authoritative data for cs.vu.nl
cs.vu.nl.      86400  IN  SOA  star boss (952771,7200,7200,2419200,86400)
cs.vu.nl.      86400  IN  TXT  "Divisie Wiskunde en Informatica."
cs.vu.nl.      86400  IN  TXT  "Vrije Universiteit Amsterdam."
cs.vu.nl.      86400  IN  MX   1 zephyr.cs.vu.nl.
cs.vu.nl.      86400  IN  MX   2 top.cs.vu.nl.

flits.cs.vu.nl. 86400  IN  HINFO Sun Unix
flits.cs.vu.nl. 86400  IN  A    130.37.16.112
flits.cs.vu.nl. 86400  IN  A    192.31.231.165
flits.cs.vu.nl. 86400  IN  MX   1 flits.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX   2 zephyr.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX   3 top.cs.vu.nl.
www.cs.vu.nl.   86400  IN  CNAME star.cs.vu.nl
ftp.cs.vu.nl.   86400  IN  CNAME zephyr.cs.vu.nl

rowboat        IN  A    130.37.56.201
               IN  MX   1 rowboat
               IN  MX   2 zephyr
               IN  HINFO Sun Unix

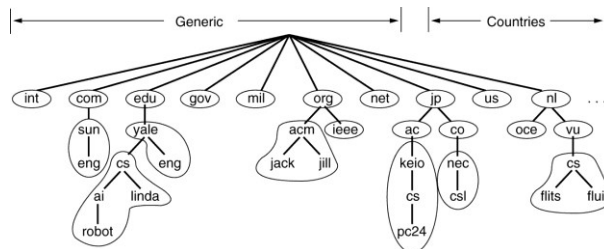
little-sister  IN  A    130.37.62.23
               IN  HINFO Mac MacOS

laserjet       IN  A    192.31.231.216
               IN  HINFO "HP Laserjet IIISI" Proprietary
```

A portion of a possible DNS database for *cs.vu.nl*.

Name Servers

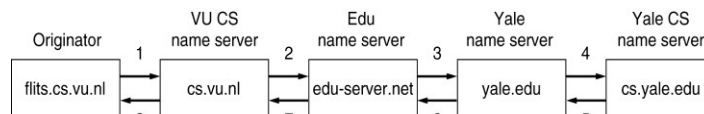
- The DNS name space is divided in non-overlapping zones
- When a resolver has a query for a domain name the local name server is queried
- If domain belongs to the zone of name server (e.g. ai.cs.yale.edu is included in cs.yale.edu):
 - authoritative resource record is replied
- Else if the domain is remote and no local information available:
 - Name server queries TLD server for the requested domain



Part of the DNS name space showing the division into zones

Name Servers

- Example: a resolver on *flits.cs.vu.nl* wants the IP of *linda.cs.yale.edu*
- The local name server (*cs.vu.nl*) is queried
- If the local name server doesn't know anything about this domain it sends an UDP packet to the TLD for edu (*edu-server.net*) that is stored in its database
- edu-server.net* must know its children so it forward the request to the name server for *yale.edu*
- yale.edu* sends the request to *cs.yale.edu* that must have the authoritative resource record
- The resource record is sent back to each client from its server
- Record delivered to *cs.vu.nl* are cached for further use (TTL specifies how long must be kept)



How a resolver looks up a remote name in eight steps.

DNS Servers – DNS Resolvers

- **DNS Servers**

- A DNS server is software that serves DNS requests for clients. It can be a cache (see above) which doesn't have any names of its own and just performs recursive queries (and caching), or it can be a "real" server, meaning that it does hold the authoritative answers for certain resources.

- **DNS Resolvers**

- DNS resolvers are just DNS clients. They can make two main types of queries: iterative, and recursive.

Recursive vs. Iterative Queries

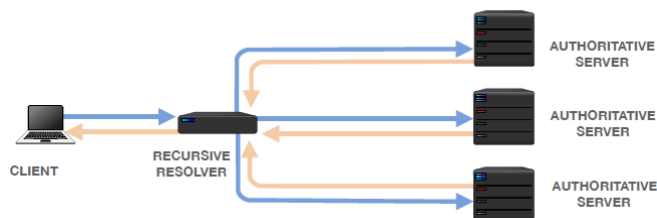
- As mentioned above, recursive queries are queries where the client asks the server to do all the work for it. It sends in its query the RECURSION DESIRED flag, and the DNS server will either honor that or not.
- Iterative queries are the opposite of recursive queries. When they're used, the server doesn't go find the answer for the client (unless it's on the first question and response), but rather tells the client where to look next. So if the client asks for chat.google.com, it tells the client to check with the .com servers and considers its work done.

Response and Queries

- **Authoritative vs. Non-authoritative Responses**
 - Authoritative responses are responses that come directly from a nameserver that has authority over the record in question. Non-authoritative answers come second-hand (or more), i.e., from another server or through a cache.
- **Reverse vs. Forward Queries**
 - Reverse queries simply reverse the direction of DNS lookups, i.e. going from IP to name instead of name to IP. Forward queries are another name for normal name-to-IP queries

Authoritative Server

- Contains records in its zone file
- Only answer queries for data under its authority
- If can't answer, it points to authority
 - but doesn't query recursively.



DNS Caches

- A DNS cache can mean a couple of different things, which is why it's confusing.
 1. The list of names and IPs that you've resolved recently, which are "cached" such that if you ask the question again you'll get the same answer without generating network traffic.
 2. A DNS server that doesn't have any authoritative names itself, but just performs recursive queries and caching (saving those answers for future requests within a certain amount of time).
- So when someone says they need to clear their DNS cache, they're probably talking about their local cache. If they're talking about setting up a DNS Cache, they're probably talking about a DNS server that just makes DNS queries faster for the network

Zone Transfers

- Zone Transfers are the means by which slave servers pull records from master servers for backup and redundancy purposes. They take place over TCP because the data being transferred is usually substantial (and mostly likely over 512-bytes). During the operation, the client sends a query type of IXFR (Incremental Zone Transfer) instead of AXFR (Normal Zone Transfer).
- Zone Transfers are sensitive from a security standpoint because when someone knows what and where your resources are, it helps them plan an attack against you. Zone Transfers should only be allowed by approved systems.

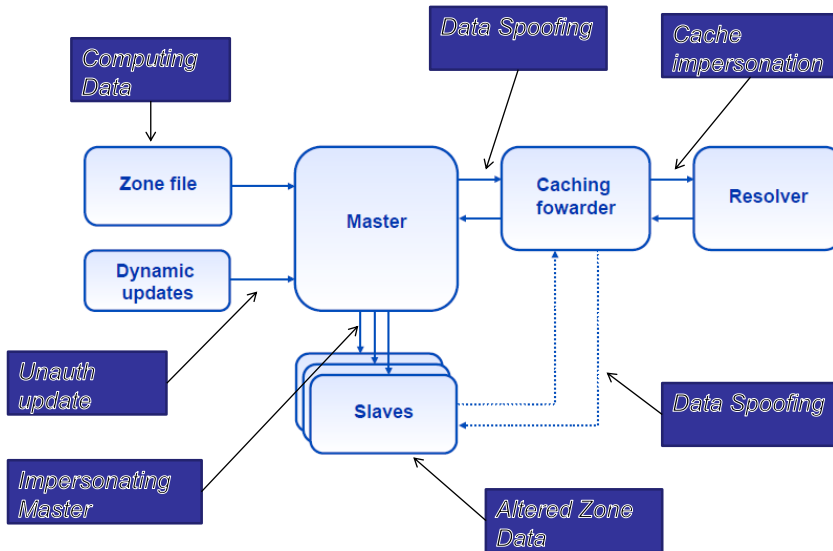
What does DNS affect ?

- DNS also helps mail delivery agents deliver mail along the Internet.
 - To deliver mail across the Internet, DNS uses **mail exchange records** (MX records)
- There is a very specific scenario in which changing your DNS servers may indirectly help to reduce lag.
 - changing your DNS servers **MAY** be able improve your upload speeds to the point where you no longer lag (**gaming, streaming**)

Basic DNS problems

- DNS is plain text
- Simple UDP, no sessions
- Tree structure with delegations
- Each entity is responsible for a limited part of it
- Resolvers victims of attacks, hijacks and mistakes
- Trust is needed

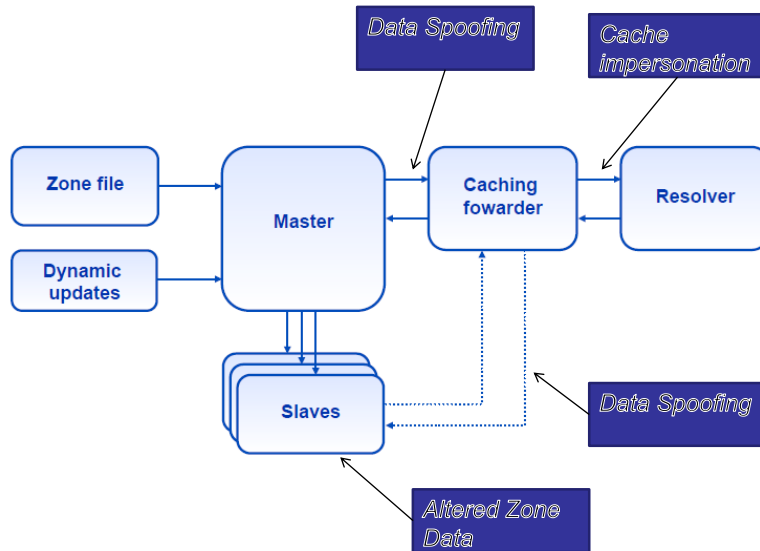
DNS Data Flow e Problems



Security for DNS: DNSSEC

- DNS Security Extensions
- RFC4033
- Adds layers on top of DNS to make it verifiable
 - Adds new record types
 - Adds PKI
- Chain of trust to validate data

DNSSEC Protected Vuln.



DNSSEC in summary

- Data authenticity and integrity by signing the Resource Records Sets with private DNSKEY
- You need Public DNSKEYs to verify the RRSIGs
- Children sign their zones with their private key
 - Parent guarantees authenticity of child's key by signing the hash of it (DS)
- Repeat for parent ...
 - ...and grandparent
- Ideal case: one public DNSKEY distributed

DNS Software

- DNS BIND – authoritative + recursive server
- Unbound - caching DNS resolver
- NSD – authoritative only nameserver
- Microsoft DNS – provided with the Windows Server
- Knot DNS – authoritative only nameserver
- PowerDNS – data storage backends

BIND

- Berkeley Internet Name Domain
- The most widely-used open source DNS software on the Internet
- Current version is Bind 9.11.4-P2
- Maintained by the Internet Systems Consortium (ISC)